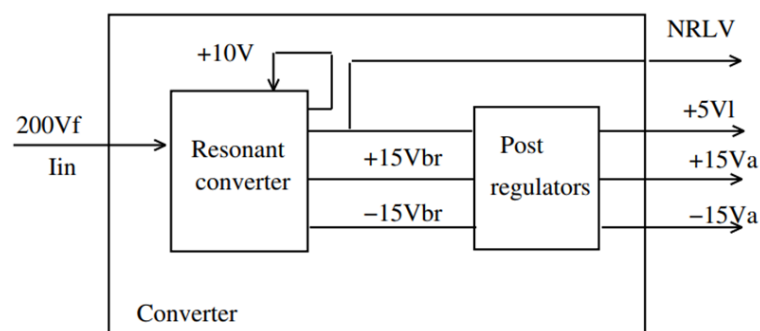


Project Title : Failure Analysis Schlumberger Internship

Student Names : Pushpal Das, RA1911004010565

Project Guide : Ms. R. Monika

ABSTRACT: As electronic devices become more complex and ubiquitous in our daily lives, the reliability and performance of printed circuit boards (PCBs) and printed wiring assemblies (PWAs) become increasingly important. Failure analysis of PCBs and PWAs is a critical aspect of ensuring their reliability, and it involves the systematic investigation of the causes of failures that occur during design, manufacturing, or operation. There are various failure mechanisms that can occur in PCBs and PWAs. Component failures are common, and they can be caused by factors such as temperature, humidity, and mechanical stress. Solder joint failures are another common failure mechanism, and they can result from poor soldering quality, thermal cycling, or mechanical stress. Trace and via failures can also occur due to issues such as manufacturing defects, electrostatic discharge, and corrosion. To conduct a failure analysis, various tools and techniques are used, depending on the nature and extent of the failure. Visual inspection is often the first step, and it involves examining the failed component or assembly for any visible signs of damage or defects. X-ray imaging is another useful technique for identifying defects in PCBs and PWAs, particularly for identifying issues with internal components, such as BGA packages. Micro sectioning is also commonly used to investigate the microstructure of solder joints or other components, and it can help identify the root cause of a failure. Once the root cause of a failure has been identified, steps can be taken to mitigate the risk of similar failures occurring in the future. This may involve changes to the design, materials, or manufacturing processes, or it may involve changes to the testing and inspection protocols to ensure that potential issues are caught before the device is released into the field. In conclusion, failure analysis of PCBs and PWAs is critical for ensuring the reliability and performance of electronic devices. By understanding the various failure mechanisms and using appropriate tools and techniques to investigate them, engineers can identify the root cause of a failure and take steps to prevent similar issues from occurring.



Objectives	Steps involved in troubleshooting any PCB or PWA, including identifying the issue, developing a hypothesis, testing, and validating the solution
Knowledge acquired in the listed courses	1. Linear Integrated Circuit (18ECC202J) 2. Digital Signal Processing (18ECC204J) 3. Analog Electronics Circuits (18ECC201J)

	4. Digital Electronics Principles (18ECC103J) 5. Electronic Devices (18ECC102J)
Realistic Constraints	1.The failure analysis must be conducted in a safe manner, particularly when analysing components that may contain hazardous materials or energy sources. In some cases, the failure analysis may be constrained by the need to conduct non-destructive testing, particularly for components that are expensive or difficult to replace. 2.The failure analysis must be conducted using equipment that is compatible with the design and components of the PCB or PWA under analysis and may be constrained by limited access to components or assemblies, particularly for products that are in the field or have limited availability. 3.The failure analysis must be conducted within a limited budget and with a limited number of personnel and equipment.
Standards to be referred/followed	IPC-7095 D: Design and Assembly Process Implementation for BGAs - This standard provides guidelines for design, assembly, and rework of Ball Grid Array (BGA) components, including failure analysis techniques. IPC-7711/21B: Rework, Modification and Repair of Electronic Assemblies - This standard provides procedures for the rework, modification, and repair of electronic assemblies, including guidelines for failure analysis.
Multidisciplinary tasks involved	Electrical Engineering: Electrical engineers are responsible for analysing the electrical properties of the board, including the performance of the components and the integrity of the electrical signals. Reliability Engineering: Reliability engineers are responsible for analysing the reliability and failure modes of the board, including the effects of aging and the probability of failure under different conditions. Quality Control and Assurance: Quality control and assurance specialists are responsible for ensuring that the board meets the required quality standards, and for identifying and addressing any defects or issues that arise during the manufacturing process or in the field.
Deliverables/Outcomes	The user can identify how the problems occurred to the board being troubleshooted and can use it for a sustaining solution for future scope.